Living Village - Assignment 5: Beta Build

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AI for Visual Computing | Beta release notes and revised schedule | November 9, 2026

Playable build: https://ishashah2303.github.io/LivingVillage/
Public source: https://github.com/ishashah2303/LivingVillage

Choose **Rule-based only** to run entirely in the browser. The optional hosted planner is not needed for the beta demo; cold starts or connection failures leave the local simulation running.

1. Primary pillar recap and secondary-pillar integration

NPC AI / behavior trees: the alpha established eight named residents with routines, needs, social relationships and episodic memory. A reactive tree prioritizes emergencies over helping friends, events and ordinary work. The inspector exposes the live tree, goal and decision source. These are prior achievements, retained in the beta.

Navigation affects the game: fires and edited obstacles update the Recast/Detour TileCache; villagers move as crowd agents. Immediate flee/guard behaviors act while a slower planner coordinates destinations. Accepted commands feed the same behaviors and movement system. In Fire Season this changes smoke exposure, assistance and firefighting, which affect injuries, village integrity and the outcome. Fire-reactive boids provide a visible response in the same scene.

Beta example: if a second fire makes an assigned refuge unsafe, the active evacuation now rechecks destination safety and reachability when the world changes, discards the invalid target and chooses a reachable safe refuge. If none exists, the villager reports that no safe route is available. Exhausted movement retries no longer count as arrival; safety timing requires physical arrival outside the danger threshold.

2. Release notes - changes since Assignment 4

Playable crowd scaling. Sandbox presets restart with 8, 16, 32 or 64 real residents, each with behavior, needs, relationships, memory and navigation. DetourCrowd capacity is 64. Presets fix seed 5, 40 birds and local planning for repeatable comparisons. Normal Fire Season remains at eight residents. Reports over 32 agents stay local to respect the server schema.

Measurement tools. Ten-second browser captures follow a three-second warm-up and export FPS, p95 frame/simulation time, draw calls and resolution. Active-game captures record changing obstacle counts. An opt-in headless sweep exercises four populations, two flock sizes and three seeds with real WASM navigation, 20 rocks and a fire.

Correctness and presentation. Added regressions for newly unsafe targets, unreachable refuges, false arrival after stuck recovery and unsafe arrival positions. Above 16 residents, only the selected name/speech label is shown; the scrollable inspector still selects every resident. Path diagnostics default off. Default birds are now 40. Help text correctly states that fires near houses cause triple integrity damage.

3. Core loop and demo

Start: choose Fire Season or Sandbox and planner. **Active:** water fires, leave food and ring the bell through escalating fire waves and a shortage. **End:** survive four simulated minutes for victory, or lose at zero integrity / three injured residents; inspect results, replay or return to the menu. Sandbox has an explicit session end. Try Fire near Maya, Stale plan and the crowd presets to inspect the integration.

Living Village - Beta validation and production plan

4. Validation and measured performance

44 frontend tests passed (two optional exporters/benchmarks skipped in the normal suite); **18 backend tests passed**, using fake LLM responses. Type checking and the production static export passed. The game suite includes bot wins and idle-player losses for three seeds. Browser checks covered start, active fire/planning, defeat, replay, menu return, Sandbox end and 64-resident selection. Browser victory and human usability testing remain follow-up checks; no paid model evaluation was repeated.

Production browser sample	Mean FPS	p95 frame	p95 sim/frame
Beta: 8 residents, 40 birds, active fire/nav updates	60.0	21.1 ms	0.80 ms
Earlier scaling pass: 16 residents / 40 birds	60.0	19.0 ms	0.50 ms
Earlier scaling pass: 32 residents / 40 birds	57.3	25.3 ms	0.80 ms
Earlier scaling pass: 64 residents / 40 birds	36.9	39.5 ms	1.70 ms

Windows, Intel i7-8550U, Chromium 156; 698 x 474 CSS-pixel canvas, DPR 1, 1x speed, spatial grid. Each sample is 10 seconds. The beta sample had 600 frames, 75-76 navigation obstacles and zero frames over 33 ms. Earlier crowd samples were morning routines with paths on, before label culling; they are historical limits, not a controlled before/after comparison. Raw JSON and conditions are in docs/benchmarks and docs/scaling-notes.md.

The eight-resident sample meets the recorded 60 FPS average target at this viewport, but p95 exceeds the 16.7 ms frame budget. Larger crowds do not establish sustained 60 FPS. The original Assignment 2 document is absent; its 8-resident / 40-bird / runtime-navigation target is quoted in the Assignment 3 report.

5. Updated schedule (Weeks 10-15)

Week / status	Engineering and evaluation	Visuals / milestone
10 | Nov 2-8 Delivered	Real crowd presets and repeatable browser/CPU benchmarks. Instancing and min-cost flow deferred.	Retain procedural village; establish measured crowd limits.
11 | Nov 9-15 Beta / follow-up	Delivered destination revalidation and truthful arrivals; regression tests, static build and report. Next: browser victory check.	Delivered label culling and clearer instructions; package beta.
12 | Nov 16-22 Planned	Human playtest; prioritize smoke-aware routes, refuge capacity and in-flight planner fallback.	Improve danger feedback and crowd readability from feedback.
13 | Nov 23-29 Planned	Controlled safety scenarios, balance and larger-viewport performance; optimize measured bottlenecks.	Review accessibility and visual consistency.
14 | Nov 30-Dec 6 Planned	Feature freeze, full regression and release-candidate benchmark.	Final screenshots and demo recording.
15 | Dec 7-13 Buffer	Bug fixes, documentation and final submission preparation; no new features.	Polish and final report.

6. Known issues and scope decisions

Routes avoid carved fire footprints but do not yet price smoke exposure along the full path. Refuge assignments are not global occupancy management. Larger crowds share six homes and repeat guard roles, so they are capacity experiments, not equal-difficulty gameplay. Individual character meshes still incur high draw counts; 64 remains a stress case.

An in-flight planner disconnection can drop that request while reflexes continue. Balance has no human playtest evidence yet. Rigged characters, audio, visible victory rain, extra hazard types, min-cost flow, persistent vector memory and a model cascade are deferred. This beta expands and stabilizes the existing pillars; it is not a feature freeze or completion of the entire earlier roadmap.